Soft constraint | NRMSE summary (util.py: RMS normalization)

Median of per-IC NRMSE on the same 0 complete test ICs for every model

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
1 ANN10	**baseline**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2/18
1 ANN10	**soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
2 P-only DeepONet10	**baseline**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
2 P-only DeepONet10	**soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
3 ANN10 + P-only DeepONet10	**baseline**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3/18
3 ANN10 + P-only DeepONet10	**soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
4 T/P DeepONet10	**baseline**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6/18
4 T/P DeepONet10	**soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2/18
5 ANN10 + T/P DeepONet10	**baseline**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	8/18
5 ANN10 + T/P DeepONet10	**soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
6 T/P/ϕ DeepONet10	**baseline**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6/18
6 T/P/ϕ DeepONet10	**soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	12/18
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0/18

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
Common ICs: .